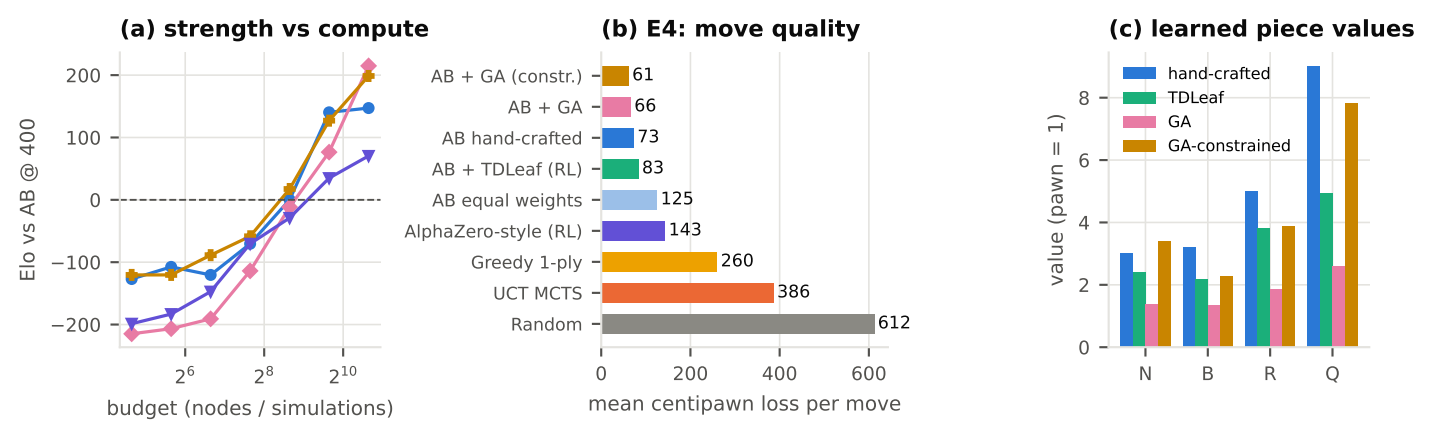
(a) strength vs compute
Elo vs AB @ 400
200
100
0
−100
−200
2^6
2^8
2^{10}
budget (nodes / simulations)
(b) E4: move quality
AB + GA (constr.)
AB + GA
AB hand-crafted
AB + TDLeaf (RL)
AB equal weights
AlphaZero-style (RL)
Greedy 1-ply
UCT MCTS
Random
61
66
73
83
125
143
260
386
612
0
200
400
600
mean centipawn loss per move
(c) learned piece values
hand-crafted
TDLeaf
GA
GA-constrained
value (pawn = 1)
0
2
4
6
8
N
B
R
Q